AN ANTI-EQUIVALENCE BETWEEN SYSTEMS OF FINITE TYPE ALGEBRAS AND FINITE TYPE MAPS OF AFFINE FORMAL ALGEBRAIC SPACES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement speaks of 'the category C introduced above', and that category is introduced in this chapter immediately before the lemma in a paragraph that is not an environment and carries no label. That paragraph is delivered as a context span, verbatim, so that the reader sees the objects of the category and its two axioms; it is not counted as a statement of the card.

Editorial note. The proof displays three computations of fibre products; all lie inside the counted span and are reproduced verbatim.

Editorial note. The counted spans carry 3 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The category of inverse systems of finite type algebras over the quotients of a complete topological ring. Same chapter, the paragraph introducing the category and its two axioms, at source lines 6394-6401, delivered as the context span. Fixes the objects of the first category and the structural condition used to identify stages.
- Finite type for a morphism of affine formal algebraic spaces. Same chapter, Definition definition-finite-type, invoked at proof lines 6420-6423 and 6451-6457. Supplies the property transferred in both directions.
- The fibre product of an affine formal algebraic space with a quotient of the ring is affine, and representable morphisms are affine. Same chapter, Lemma lemma-affine-representable-by-algebraic-spaces, invoked at proof lines 6418-6420. Makes the fibre products appearing in the construction affine so that their coordinate rings may be taken.
- Morphisms from a quasi-compact scheme into a filtered colimit factor through a stage. Same chapter, Lemma lemma-factor-through-thickening, invoked at proof lines 6436-6438 and 6451-6453. Underlies the evaluation on quasi-compact test objects used in all three verifications.

Source reference. Original definition 0AM4: sources/stacks-algebra/pinned-original-source.tex, source lines 735–735. Pointer outside the counted span, invoked once

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for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AKN: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4209–4209. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIA: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2080–2080. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Let A be a topological ring complete with respect to a linear topology. Let (I_λ) be a fundamental system of open ideals. Let $\mathcal{C}$ be the category of inverse systems (B_λ) where

(1) B_λ is a finite type A/I_λ-algebra, and
(2) $B_\mu \to B_\lambda$ is an A/I_μ-algebra homomorphism which induces an isomorphism $B_\mu/I_\lambda B_\mu \to B_\lambda$.

Lemma 0.1. *Let S be a scheme. Let X be an affine formal algebraic space over S. Assume X is McQuillan and let A be the weakly admissible topological ring associated to X. Then there is an anti-equivalence of categories between*

(1) *the category $\mathcal{C}$ introduced above, and*
(2) *the category of maps $Y \to X$ of finite type of affine formal algebraic spaces.*

Proof. Let (I_λ) be a fundamental system of weakly admissible ideals of definition in A. Consider Y as in (2). Then $Y \times_X \operatorname{Spec}(A/I_\lambda)$ is affine (Definition 0AM4 and Lemma 0AKN). Say $Y \times_X \operatorname{Spec}(A/I_\lambda) = \operatorname{Spec}(B_\lambda)$. The ring map $A/I_\lambda \to B_\lambda$ is of finite type because $\operatorname{Spec}(B_\lambda) \to \operatorname{Spec}(A/I_\lambda)$ is of finite type (by Definition 0AM4). Then (B_λ) is an object of $\mathcal{C}$.

Conversely, given an object (B_λ) of $\mathcal{C}$ we can set $Y = \operatorname{colim} \operatorname{Spec}(B_\lambda)$. This is an affine formal algebraic space. We claim that

$$Y \times_X \operatorname{Spec}(A/I_\lambda) = (\operatorname{colim}_\mu \operatorname{Spec}(B_\mu)) \times_X \operatorname{Spec}(A/I_\lambda) = \operatorname{Spec}(B_\lambda)$$

To show this it suffices we get the same values if we evaluate on a quasi-compact scheme U. A morphism $U \to (\operatorname{colim}_\mu \operatorname{Spec}(B_\mu)) \times_X \operatorname{Spec}(A/I_\lambda)$ comes from a morphism $U \to \operatorname{Spec}(B_\mu) \times_{\operatorname{Spec}(A/I_\mu)} \operatorname{Spec}(A/I_\lambda)$ for some $\mu \geq \lambda$ (use Lemma 0AIA two times). Since $\operatorname{Spec}(B_\mu) \times_{\operatorname{Spec}(A/I_\mu)} \operatorname{Spec}(A/I_\lambda) = \operatorname{Spec}(B_\lambda)$ by our second assumption on objects of $\mathcal{C}$ this proves what we want. Using this we can show the morphism $Y \to X$ is of finite type. Namely, we note that for any morphism $U \to X$ with U a quasi-compact scheme, we get a factorization $U \to \operatorname{Spec}(A/I_\lambda) \to X$ for some λ (see lemma cited above). Hence

$$Y \times_X U = Y \times_X \operatorname{Spec}(A/I_\lambda)) \times_{\operatorname{Spec}(A/I_\lambda)} U = \operatorname{Spec}(B_\lambda) \times_{\operatorname{Spec}(A/I_\lambda)} U$$

is a scheme of finite type over U as desired. Thus the construction $(B_\lambda) \mapsto \operatorname{colim} \operatorname{Spec}(B_\lambda)$ does give a functor from category (1) to category (2).

To finish the proof we show that the above constructions define quasi-inverse functors between the categories (1) and (2). In one direction you have to show that

$$(\operatorname{colim}_\mu \operatorname{Spec}(B_\mu)) \times_X \operatorname{Spec}(A/I_\lambda) = \operatorname{Spec}(B_\lambda)$$

for any object (B_λ) in the category $\mathcal{C}$. This we proved above. For the other direction you have to show that

$$Y = \operatorname{colim}(Y \times_X \operatorname{Spec}(A/I_\lambda))$$

given Y in the category (2). Again this is true by evaluating on quasi-compact test objects and because $X = \operatorname{colim} \operatorname{Spec}(A/I_\lambda)$. □